

1.

a.

$$\det A = a_{31}C_{31} + a_{35}C_{35} = 3C_{31} + (-1)C_{35} \quad (\text{along } R_3)$$

$$M_{31} = \begin{vmatrix} 4 & -2 & 4 & 6 \\ 3 & -1 & 9 & 5 \\ 3 & 4 & 3 & 1 \\ 4 & 8 & 0 & -3 \end{vmatrix} \quad (\text{along } R_4)$$

$$\begin{vmatrix} -2 & 4 & 6 \\ -1 & 9 & 5 \\ 4 & 3 & 1 \end{vmatrix} = (-2)[(9)(1) - (5)(3)] - 4[(-1)(1) - (5)(4)] + 6[(-1)(3) - (9)(4)]$$

$$= (-2)(-6) - 4(-21) + 6(-39) = -138$$

$$\begin{vmatrix} 4 & 4 & 6 \\ 3 & 9 & 5 \\ 3 & 3 & 1 \end{vmatrix} = 4[(9)(1) - (5)(3)] - 4[(3)(1) - (5)(3)] + 6[(3)(3) - (9)(3)]$$

$$= 4(-6) - 4(-12) + 6(-18) = -84$$

$$\begin{vmatrix} 4 & -2 & 4 \\ 3 & -1 & 9 \\ 3 & 4 & 3 \end{vmatrix} = 4[(-1)(3) - (9)(4)] - (-2)[(3)(3) - (9)(3)] + 4[(3)(4) - (-1)(3)]$$

$$= 4(-39) - (-2)(-18) + 4(15) = -132$$

$$M_{31} = -4(-138) + 8(-84) + (-3)(-132) = 276$$

$$C_{31} = (-1)^{3+1}(276) = 276$$

$$M_{35} = \begin{vmatrix} 1 & 4 & -2 & 4 \\ 0 & 3 & -1 & 9 \\ 0 & 3 & 4 & 3 \\ 0 & 4 & 8 & 0 \end{vmatrix} \quad (\text{along } C_1)$$

$$\begin{vmatrix} 3 & -1 & 9 \\ 3 & 4 & 3 \\ 4 & 8 & 0 \end{vmatrix} = 3[(4)(0) - (3)(8)] - (-1)[(3)(0) - (3)(4)] + 9[(3)(8) - (4)(4)]$$

$$= 3(-24) - (-1)(-12) + 9(8) = -12$$

$$M_{35} = 1(-12) = -12$$

$$C_{35} = (-1)^{3+5}(-12) = -12$$

$$\det A = 3(276) + (-1)(-12) = 828 + 12$$

$$\boxed{\det A = 840}$$

b.

$$M_{11} = \begin{vmatrix} 3 & -1 & 9 & 5 \\ 0 & 0 & 0 & -1 \\ 3 & 4 & 3 & 1 \\ 4 & 8 & 0 & -3 \end{vmatrix} \quad (\text{along } R_2)$$

$$\begin{vmatrix} 3 & -1 & 9 \\ 3 & 4 & 3 \\ 4 & 8 & 0 \end{vmatrix} = 3[(4)(0) - (3)(8)] - (-1)[(3)(0) - (3)(4)] + 9[(3)(8) - (4)(4)]$$

$$= 3(-24) - (-1)(-12) + 9(8) = -12$$

$$M_{11} = (-1)(-12) = 12$$

$$M_{12} = \begin{vmatrix} 0 & -1 & 9 & 5 \\ 3 & 0 & 0 & -1 \\ 0 & 4 & 3 & 1 \\ 0 & 8 & 0 & -3 \end{vmatrix} \quad (\text{along } C_1)$$

$$\begin{vmatrix} -1 & 9 & 5 \\ 4 & 3 & 1 \\ 8 & 0 & -3 \end{vmatrix} = (-1)[(3)(-3) - (1)(0)] - 9[(4)(-3) - (1)(8)] + 5[(4)(0) - (3)(8)]$$

$$= (-1)(-9) - 9(-20) + 5(-24) = 69$$

$$M_{12} = -3(69) = -207$$

$$M_{13} = \begin{vmatrix} 0 & 3 & 9 & 5 \\ 3 & 0 & 0 & -1 \\ 0 & 3 & 3 & 1 \\ 0 & 4 & 0 & -3 \end{vmatrix} \quad (\text{along } C_1)$$

$$\begin{vmatrix} 3 & 9 & 5 \\ 3 & 3 & 1 \\ 4 & 0 & -3 \end{vmatrix} = 3[(3)(-3) - (1)(0)] - 9[(3)(-3) - (1)(4)] + 5[(3)(0) - (3)(4)]$$

$$= 3(-9) - 9(-13) + 5(-12) = 30$$

$$M_{13} = -3(30) = -90$$

$$M_{14} = \begin{vmatrix} 0 & 3 & -1 & 5 \\ 3 & 0 & 0 & -1 \\ 0 & 3 & 4 & 1 \\ 0 & 4 & 8 & -3 \end{vmatrix} \quad (\text{along } C_1)$$

$$\begin{vmatrix} 3 & -1 & 5 \\ 3 & 4 & 1 \\ 4 & 8 & -3 \end{vmatrix} = 3[(4)(-3) - (1)(8)] - (-1)[(3)(-3) - (1)(4)] + 5[(3)(8) - (4)(4)]$$

$$= 3(-20) - (-1)(-13) + 5(8) = -33$$

$$M_{14} = -3(-33) = 99$$

$$M_{15} = \begin{vmatrix} 0 & 3 & -1 & 9 \\ 3 & 0 & 0 & 0 \\ 0 & 3 & 4 & 3 \\ 0 & 4 & 8 & 0 \end{vmatrix} \quad (\text{along } R_2)$$

$$\begin{vmatrix} 3 & -1 & 9 \\ 3 & 4 & 3 \\ 4 & 8 & 0 \end{vmatrix} = 3[(4)(0) - (3)(8)] - (-1)[(3)(0) - (3)(4)] + 9[(3)(8) - (4)(4)]$$

$$= 3(-24) - (-1)(-12) + 9(8) = -12$$

$$M_{15} = -3(-12) = 36$$

$$M_{21} = \begin{vmatrix} 4 & -2 & 4 & 6 \\ 0 & 0 & 0 & -1 \\ 3 & 4 & 3 & 1 \\ 4 & 8 & 0 & -3 \end{vmatrix} \quad (\text{along } R_2)$$

$$\begin{vmatrix} 4 & -2 & 4 \\ 3 & 4 & 3 \\ 4 & 8 & 0 \end{vmatrix} = 4[(4)(0) - (3)(8)] - (-2)[(3)(0) - (3)(4)] + 4[(3)(8) - (4)(4)]$$

$$= 4(-24) - (-2)(-12) + 4(8) = -88$$

$$M_{21} = (-1)(-88) = 88$$

$$M_{22} = \begin{vmatrix} 1 & -2 & 4 & 6 \\ 3 & 0 & 0 & -1 \\ 0 & 4 & 3 & 1 \\ 0 & 8 & 0 & -3 \end{vmatrix} \quad (\text{along } R_2)$$

$$\begin{vmatrix} -2 & 4 & 6 \\ 4 & 3 & 1 \\ 8 & 0 & -3 \end{vmatrix} = (-2)[(3)(-3) - (1)(0)] - 4[(4)(-3) - (1)(8)] + 6[(4)(0) - (3)(8)]$$

$$= (-2)(-9) - 4(-20) + 6(-24) = -46$$

$$\begin{vmatrix} 1 & -2 & 4 \\ 0 & 4 & 3 \\ 0 & 8 & 0 \end{vmatrix} = 1[(4)(0) - (3)(8)] - (-2)[(0)(0) - (3)(0)] + 4[(0)(8) - (4)(0)]$$

$$= 1(-24) - (-2)(0) + 4(0) = -24$$

$$M_{22} = -3(-46) + (-1)(-24) = 162$$

$$M_{23} = \begin{vmatrix} 1 & 4 & 4 & 6 \\ 3 & 0 & 0 & -1 \\ 0 & 3 & 3 & 1 \\ 0 & 4 & 0 & -3 \end{vmatrix} \quad (\text{along } R_2)$$

$$\begin{vmatrix} 4 & 4 & 6 \\ 3 & 3 & 1 \\ 4 & 0 & -3 \end{vmatrix} = 4[(3)(-3) - (1)(0)] - 4[(3)(-3) - (1)(4)] + 6[(3)(0) - (3)(4)]$$

$$= 4(-9) - 4(-13) + 6(-12) = -56$$

$$\begin{vmatrix} 1 & 4 & 4 \\ 0 & 3 & 3 \\ 0 & 4 & 0 \end{vmatrix} = 1[(3)(0) - (3)(4)] - 4[(0)(0) - (3)(0)] + 4[(0)(4) - (3)(0)]$$

$$= 1(-12) - 4(0) + 4(0) = -12$$

$$M_{23} = -3(-56) + (-1)(-12) = 180$$

$$M_{24} = \begin{vmatrix} 1 & 4 & -2 & 6 \\ 3 & 0 & 0 & -1 \\ 0 & 3 & 4 & 1 \\ 0 & 4 & 8 & -3 \end{vmatrix} \quad (\text{along } R_2)$$

$$\begin{vmatrix} 4 & -2 & 6 \\ 3 & 4 & 1 \\ 4 & 8 & -3 \end{vmatrix} = 4[(4)(-3) - (1)(8)] - (-2)[(3)(-3) - (1)(4)] + 6[(3)(8) - (4)(4)]$$

$$= 4(-20) - (-2)(-13) + 6(8) = -58$$

$$\begin{vmatrix} 1 & 4 & -2 \\ 0 & 3 & 4 \\ 0 & 4 & 8 \end{vmatrix} = 1[(3)(8) - (4)(4)] - 4[(0)(8) - (4)(0)] + (-2)[(0)(4) - (3)(0)]$$

$$= 1(8) - 4(0) + (-2)(0) = 8$$

$$M_{24} = -3(-58) + (-1)(8) = 166$$

$$M_{25} = \begin{vmatrix} 1 & 4 & -2 & 4 \\ 3 & 0 & 0 & 0 \\ 0 & 3 & 4 & 3 \\ 0 & 4 & 8 & 0 \end{vmatrix} \quad (\text{along } R_2)$$

$$\begin{vmatrix} 4 & -2 & 4 \\ 3 & 4 & 3 \\ 4 & 8 & 0 \end{vmatrix} = 4[(4)(0) - (3)(8)] - (-2)[(3)(0) - (3)(4)] + 4[(3)(8) - (4)(4)]$$

$$= 4(-24) - (-2)(-12) + 4(8) = -88$$

$$M_{25} = -3(-88) = 264$$

$$M_{31} = \begin{vmatrix} 4 & -2 & 4 & 6 \\ 3 & -1 & 9 & 5 \\ 3 & 4 & 3 & 1 \\ 4 & 8 & 0 & -3 \end{vmatrix} \quad (\text{along } R_4)$$

$$\begin{vmatrix} -2 & 4 & 6 \\ -1 & 9 & 5 \\ 4 & 3 & 1 \end{vmatrix} = (-2)[(9)(1) - (5)(3)] - 4[(-1)(1) - (5)(4)] + 6[(-1)(3) - (9)(4)]$$

$$= (-2)(-6) - 4(-21) + 6(-39) = -138$$

$$\begin{vmatrix} 4 & 4 & 6 \\ 3 & 9 & 5 \\ 3 & 3 & 1 \end{vmatrix} = 4[(9)(1) - (5)(3)] - 4[(3)(1) - (5)(3)] + 6[(3)(3) - (9)(3)]$$

$$= 4(-6) - 4(-12) + 6(-18) = -84$$

$$\begin{vmatrix} 4 & -2 & 4 \\ 3 & -1 & 9 \\ 3 & 4 & 3 \end{vmatrix} = 4[(-1)(3) - (9)(4)] - (-2)[(3)(3) - (9)(3)] + 4[(3)(4) - (-1)(3)]$$

$$= 4(-39) - (-2)(-18) + 4(15) = -132$$

$$M_{31} = -4(-138) + 8(-84) + (-3)(-132) = 276$$

$$M_{32} = \begin{vmatrix} 1 & -2 & 4 & 6 \\ 0 & -1 & 9 & 5 \\ 0 & 4 & 3 & 1 \\ 0 & 8 & 0 & -3 \end{vmatrix} \quad (\text{along } C_1)$$

$$\begin{vmatrix} -1 & 9 & 5 \\ 4 & 3 & 1 \\ 8 & 0 & -3 \end{vmatrix} = (-1)[(3)(-3) - (1)(0)] - 9[(4)(-3) - (1)(8)] + 5[(4)(0) - (3)(8)]$$

$$= (-1)(-9) - 9(-20) + 5(-24) = 69$$

$$M_{32} = 1(69) = 69$$

$$M_{33} = \begin{vmatrix} 1 & 4 & 4 & 6 \\ 0 & 3 & 9 & 5 \\ 0 & 3 & 3 & 1 \\ 0 & 4 & 0 & -3 \end{vmatrix} \quad (\text{along } C_1)$$

$$\begin{vmatrix} 3 & 9 & 5 \\ 3 & 3 & 1 \\ 4 & 0 & -3 \end{vmatrix} = 3[(3)(-3) - (1)(0)] - 9[(3)(-3) - (1)(4)] + 5[(3)(0) - (3)(4)]$$

$$= 3(-9) - 9(-13) + 5(-12) = 30$$

$$M_{33} = 1(30) = 30$$

$$M_{34} = \begin{vmatrix} 1 & 4 & -2 & 6 \\ 0 & 3 & -1 & 5 \\ 0 & 3 & 4 & 1 \\ 0 & 4 & 8 & -3 \end{vmatrix} \quad (\text{along } C_1)$$

$$\begin{vmatrix} 3 & -1 & 5 \\ 3 & 4 & 1 \\ 4 & 8 & -3 \end{vmatrix} = 3[(4)(-3) - (1)(8)] - (-1)[(3)(-3) - (1)(4)] + 5[(3)(8) - (4)(4)]$$

$$= 3(-20) - (-1)(-13) + 5(8) = -33$$

$$M_{34} = 1(-33) = -33$$

$$M_{35} = \begin{vmatrix} 1 & 4 & -2 & 4 \\ 0 & 3 & -1 & 9 \\ 0 & 3 & 4 & 3 \\ 0 & 4 & 8 & 0 \end{vmatrix} \quad (\text{along } C_1)$$

$$\begin{vmatrix} 3 & -1 & 9 \\ 3 & 4 & 3 \\ 4 & 8 & 0 \end{vmatrix} = 3[(4)(0) - (3)(8)] - (-1)[(3)(0) - (3)(4)] + 9[(3)(8) - (4)(4)]$$

$$= 3(-24) - (-1)(-12) + 9(8) = -12$$

$$M_{35} = 1(-12) = -12$$

$$M_{41} = \begin{vmatrix} 4 & -2 & 4 & 6 \\ 3 & -1 & 9 & 5 \\ 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & -3 \end{vmatrix} \quad (\text{along } R_3)$$

$$\begin{vmatrix} 4 & -2 & 4 \\ 3 & -1 & 9 \\ 4 & 8 & 0 \end{vmatrix} = 4[(-1)(0) - (9)(8)] - (-2)[(3)(0) - (9)(4)] + 4[(3)(8) - (-1)(4)]$$

$$= 4(-72) - (-2)(-36) + 4(28) = -248$$

$$M_{41} = -(-1)(-248) = -248$$

$$M_{42} = \begin{vmatrix} 1 & -2 & 4 & 6 \\ 0 & -1 & 9 & 5 \\ 3 & 0 & 0 & -1 \\ 0 & 8 & 0 & -3 \end{vmatrix} \quad (\text{along } R_3)$$

$$\begin{vmatrix} -2 & 4 & 6 \\ -1 & 9 & 5 \\ 8 & 0 & -3 \end{vmatrix} = (-2)[(9)(-3) - (5)(0)] - 4[(-1)(-3) - (5)(8)] + 6[(-1)(0) - (9)(8)]$$

$$= (-2)(-27) - 4(-37) + 6(-72) = -230$$

$$\begin{vmatrix} 1 & -2 & 4 \\ 0 & -1 & 9 \\ 0 & 8 & 0 \end{vmatrix} = 1[(-1)(0) - (9)(8)] - (-2)[(0)(0) - (9)(0)] + 4[(0)(8) - (-1)(0)]$$

$$= 1(-72) - (-2)(0) + 4(0) = -72$$

$$M_{42} = 3(-230) - (-1)(-72) = -762$$

$$M_{43} = \begin{vmatrix} 1 & 4 & 4 & 6 \\ 0 & 3 & 9 & 5 \\ 3 & 0 & 0 & -1 \\ 0 & 4 & 0 & -3 \end{vmatrix} \quad (\text{along } R_3)$$

$$\begin{vmatrix} 4 & 4 & 6 \\ 3 & 9 & 5 \\ 4 & 0 & -3 \end{vmatrix} = 4[(9)(-3) - (5)(0)] - 4[(3)(-3) - (5)(4)] + 6[(3)(0) - (9)(4)]$$

$$= 4(-27) - 4(-29) + 6(-36) = -208$$

$$\begin{vmatrix} 1 & 4 & 4 \\ 0 & 3 & 9 \\ 0 & 4 & 0 \end{vmatrix} = 1[(3)(0) - (9)(4)] - 4[(0)(0) - (9)(0)] + 4[(0)(4) - (3)(0)]$$

$$= 1(-36) - 4(0) + 4(0) = -36$$

$$M_{43} = 3(-208) - (-1)(-36) = -660$$

$$M_{44} = \begin{vmatrix} 1 & 4 & -2 & 6 \\ 0 & 3 & -1 & 5 \\ 3 & 0 & 0 & -1 \\ 0 & 4 & 8 & -3 \end{vmatrix} \quad (\text{along } R_3)$$

$$\begin{vmatrix} 4 & -2 & 6 \\ 3 & -1 & 5 \\ 4 & 8 & -3 \end{vmatrix} = 4[(-1)(-3) - (5)(8)] - (-2)[(3)(-3) - (5)(4)] + 6[(3)(8) - (-1)(4)]$$

$$= 4(-37) - (-2)(-29) + 6(28) = -38$$

$$\begin{vmatrix} 1 & 4 & -2 \\ 0 & 3 & -1 \\ 0 & 4 & 8 \end{vmatrix} = 1[(3)(8) - (-1)(4)] - 4[(0)(8) - (-1)(0)] + (-2)[(0)(4) - (3)(0)]$$

$$= 1(28) - 4(0) + (-2)(0) = 28$$

$$M_{44} = 3(-38) - (-1)(28) = -86$$

$$M_{45} = \begin{vmatrix} 1 & 4 & -2 & 4 \\ 0 & 3 & -1 & 9 \\ 3 & 0 & 0 & 0 \\ 0 & 4 & 8 & 0 \end{vmatrix} \quad (\text{along } R_3)$$

$$\begin{vmatrix} 4 & -2 & 4 \\ 3 & -1 & 9 \\ 4 & 8 & 0 \end{vmatrix} = 4[(-1)(0) - (9)(8)] - (-2)[(3)(0) - (9)(4)] + 4[(3)(8) - (-1)(4)]$$

$$= 4(-72) - (-2)(-36) + 4(28) = -248$$

$$M_{45} = 3(-248) = -744$$

$$M_{51} = \begin{vmatrix} 4 & -2 & 4 & 6 \\ 3 & -1 & 9 & 5 \\ 0 & 0 & 0 & -1 \\ 3 & 4 & 3 & 1 \end{vmatrix} \quad (\text{along } R_3)$$

$$\begin{vmatrix} 4 & -2 & 4 \\ 3 & -1 & 9 \\ 3 & 4 & 3 \end{vmatrix} = 4[(-1)(3) - (9)(4)] - (-2)[(3)(3) - (9)(3)] + 4[(3)(4) - (-1)(3)]$$

$$= 4(-39) - (-2)(-18) + 4(15) = -132$$

$$M_{51} = -(-1)(-132) = -132$$

$$M_{52} = \begin{vmatrix} 1 & -2 & 4 & 6 \\ 0 & -1 & 9 & 5 \\ 3 & 0 & 0 & -1 \\ 0 & 4 & 3 & 1 \end{vmatrix} \quad (\text{along } R_3)$$

$$\begin{vmatrix} -2 & 4 & 6 \\ -1 & 9 & 5 \\ 4 & 3 & 1 \end{vmatrix} = (-2)[(9)(1) - (5)(3)] - 4[(-1)(1) - (5)(4)] + 6[(-1)(3) - (9)(4)]$$

$$= (-2)(-6) - 4(-21) + 6(-39) = -138$$

$$\begin{vmatrix} 1 & -2 & 4 \\ 0 & -1 & 9 \\ 0 & 4 & 3 \end{vmatrix} = 1[(-1)(3) - (9)(4)] - (-2)[(0)(3) - (9)(0)] + 4[(0)(4) - (-1)(0)]$$

$$= 1(-39) - (-2)(0) + 4(0) = -39$$

$$M_{52} = 3(-138) - (-1)(-39) = -453$$

$$M_{53} = \begin{vmatrix} 1 & 4 & 4 & 6 \\ 0 & 3 & 9 & 5 \\ 3 & 0 & 0 & -1 \\ 0 & 3 & 3 & 1 \end{vmatrix} \quad (\text{along } R_3)$$

$$\begin{vmatrix} 4 & 4 & 6 \\ 3 & 9 & 5 \\ 3 & 3 & 1 \end{vmatrix} = 4[(9)(1) - (5)(3)] - 4[(3)(1) - (5)(3)] + 6[(3)(3) - (9)(3)]$$

$$= 4(-6) - 4(-12) + 6(-18) = -84$$

$$\begin{vmatrix} 1 & 4 & 4 \\ 0 & 3 & 9 \\ 0 & 3 & 3 \end{vmatrix} = 1[(3)(3) - (9)(3)] - 4[(0)(3) - (9)(0)] + 4[(0)(3) - (3)(0)]$$

$$= 1(-18) - 4(0) + 4(0) = -18$$

$$M_{53} = 3(-84) - (-1)(-18) = -270$$

$$M_{54} = \begin{vmatrix} 1 & 4 & -2 & 6 \\ 0 & 3 & -1 & 5 \\ 3 & 0 & 0 & -1 \\ 0 & 3 & 4 & 1 \end{vmatrix} \quad (\text{along } R_3)$$

$$\begin{vmatrix} 4 & -2 & 6 \\ 3 & -1 & 5 \\ 3 & 4 & 1 \end{vmatrix} = 4[(-1)(1) - (5)(4)] - (-2)[(3)(1) - (5)(3)] + 6[(3)(4) - (-1)(3)]$$

$$= 4(-21) - (-2)(-12) + 6(15) = -18$$

$$\begin{vmatrix} 1 & 4 & -2 \\ 0 & 3 & -1 \\ 0 & 3 & 4 \end{vmatrix} = 1[(3)(4) - (-1)(3)] - 4[(0)(4) - (-1)(0)] + (-2)[(0)(3) - (3)(0)]$$

$$= 1(15) - 4(0) + (-2)(0) = 15$$

$$M_{54} = 3(-18) - (-1)(15) = -39$$

$$M_{55} = \begin{vmatrix} 1 & 4 & -2 & 4 \\ 0 & 3 & -1 & 9 \\ 3 & 0 & 0 & 0 \\ 0 & 3 & 4 & 3 \end{vmatrix} \quad (\text{along } R_3)$$

$$\begin{vmatrix} 4 & -2 & 4 \\ 3 & -1 & 9 \\ 3 & 4 & 3 \end{vmatrix} = 4[(-1)(3) - (9)(4)] - (-2)[(3)(3) - (9)(3)] + 4[(3)(4) - (-1)(3)]$$

$$= 4(-39) - (-2)(-18) + 4(15) = -132$$

$$M_{55} = 3(-132) = -396$$

$$M = \begin{bmatrix} 12 & -207 & -90 & 99 & 36 \\ 88 & 162 & 180 & 166 & 264 \\ 276 & 69 & 30 & -33 & -12 \\ -248 & -762 & -660 & -86 & -744 \\ -132 & -453 & -270 & -39 & -396 \end{bmatrix}$$

c.

$$C_{ij} = (-1)^{i+j} M_{ij}$$

$$C_{11} = + (12) = 12, \quad C_{12} = -(-207) = 207, \quad C_{13} = +(-90) = -90, \quad C_{14} = -(99) = -99, \quad C_{15} = +(36) = 36$$

$$C_{21} = -(88) = -88, \quad C_{22} = +(162) = 162, \quad C_{23} = -(180) = -180, \quad C_{24} = +(166) = 166, \quad C_{25} = -(264) = -264$$

$$C_{31} = +(276) = 276, \quad C_{32} = -(69) = -69, \quad C_{33} = +(30) = 30, \quad C_{34} = -(-33) = 33, \quad C_{35} = +(-12) = -12$$

$$C_{41} = -(-248) = 248, \quad C_{42} = +(-762) = -762, \quad C_{43} = -(-660) = 660, \quad C_{44} = +(-86) = -86, \quad C_{45} = -(-744) = 744$$

$$C_{51} = +(-132) = -132, \quad C_{52} = -(-453) = 453, \quad C_{53} = +(-270) = -270, \quad C_{54} = -(-39) = 39, \quad C_{55} = +(-396) = -396$$

$$C = \begin{bmatrix} 12 & 207 & -90 & -99 & 36 \\ -88 & 162 & -180 & 166 & -264 \\ 276 & -69 & 30 & 33 & -12 \\ 248 & -762 & 660 & -86 & 744 \\ -132 & 453 & -270 & 39 & -396 \end{bmatrix}$$

d.

$$\text{adj } A = C^T = \begin{bmatrix} 12 & -88 & 276 & 248 & -132 \\ 207 & 162 & -69 & -762 & 453 \\ -90 & -180 & 30 & 660 & -270 \\ -99 & 166 & 33 & -86 & 39 \\ 36 & -264 & -12 & 744 & -396 \end{bmatrix}$$

e.

$$R_3 \rightarrow R_3 - 3R_1 \quad \rightarrow R_2 \rightarrow \frac{1}{3}R_2$$

$$= [0 \quad -12 \quad 6 \quad -12 \quad -19 \mid -3 \quad 0 \quad 1 \quad 0 \quad 0] = [0 \quad 1 \quad -\frac{1}{3} \quad 3 \quad \frac{5}{3} \mid 0 \quad \frac{1}{3} \quad 0 \quad 0 \quad 0]$$

$$\rightarrow R_1 \rightarrow R_1 - 4R_2 \quad \rightarrow R_3 \rightarrow R_3 + 12R_2 \quad \rightarrow R_4 \rightarrow R_4 - 3R_2$$

$$= [1 \quad 0 \quad -\frac{2}{3} \quad -8 \quad -\frac{2}{3} \mid 1 \quad -\frac{4}{3} \quad 0 \quad 0 \quad 0] = [0 \quad 0 \quad 2 \quad 24 \quad 1 \mid -3 \quad 4 \quad 1 \quad 0 \quad 0] = [0 \quad 0 \quad 5 \quad -6 \quad -4 \mid 0 \quad -1 \quad 0 \quad 1 \quad 0]$$

$$\begin{aligned}
& \rightarrow R_5 \rightarrow R_5 - 4R_2 & \rightarrow R_3 \rightarrow \frac{1}{2}R_3 & \rightarrow R_1 \rightarrow R_1 + \frac{2}{3}R_3 \\
& = [0 \ 0 \ \frac{28}{3} \ -12 \ -\frac{29}{3} \ | \ 0 \ -\frac{4}{3} \ 0 \ 0 \ 1] = [0 \ 0 \ 1 \ 12 \ \frac{1}{2} \ | \ -\frac{3}{2} \ 2 \ \frac{1}{2} \ 0 \ 0] = [1 \ 0 \ 0 \ 0 \ -\frac{1}{3} \ | \ 0 \ 0 \ \frac{1}{3} \ 0 \ 0] \\
& \rightarrow R_2 \rightarrow R_2 + \frac{1}{3}R_3 & \rightarrow R_4 \rightarrow R_4 - 5R_3 \\
& = [0 \ 1 \ 0 \ 7 \ \frac{11}{6} \ | \ -\frac{1}{2} \ 1 \ \frac{1}{6} \ 0 \ 0] = [0 \ 0 \ 0 \ -66 \ -\frac{13}{2} \ | \ \frac{15}{2} \ -11 \ -\frac{5}{2} \ 1 \ 0] \\
& \rightarrow R_5 \rightarrow R_5 - \frac{28}{3}R_3 & \rightarrow R_4 \rightarrow -\frac{1}{66}R_4 \\
& = [0 \ 0 \ 0 \ -124 \ -\frac{43}{3} \ | \ 14 \ -20 \ -\frac{14}{3} \ 0 \ 1] = [0 \ 0 \ 0 \ 1 \ \frac{13}{132} \ | \ -\frac{5}{44} \ \frac{1}{6} \ \frac{5}{132} \ -\frac{1}{66} \ 0] \\
& \rightarrow R_2 \rightarrow R_2 - 7R_4 & \rightarrow R_3 \rightarrow R_3 - 12R_4 \\
& = [0 \ 1 \ 0 \ 0 \ \frac{151}{132} \ | \ \frac{13}{44} \ -\frac{1}{6} \ -\frac{13}{132} \ \frac{7}{66} \ 0] = [0 \ 0 \ 1 \ 0 \ -\frac{15}{22} \ | \ -\frac{3}{22} \ 0 \ \frac{1}{22} \ \frac{2}{11} \ 0] \\
& \rightarrow R_5 \rightarrow R_5 + 124R_4 & \rightarrow R_5 \rightarrow -\frac{33}{70}R_5 \\
& = [0 \ 0 \ 0 \ 0 \ -\frac{70}{33} \ | \ -\frac{1}{11} \ \frac{2}{3} \ \frac{1}{33} \ -\frac{62}{33} \ 1] = [0 \ 0 \ 0 \ 0 \ 1 \ | \ \frac{3}{70} \ -\frac{11}{35} \ -\frac{1}{70} \ \frac{31}{35} \ -\frac{33}{70}] \\
& \rightarrow R_1 \rightarrow R_1 + \frac{1}{3}R_5 & \rightarrow R_2 \rightarrow R_2 - \frac{151}{132}R_5 \\
& = [1 \ 0 \ 0 \ 0 \ 0 \ | \ \frac{1}{70} \ -\frac{11}{105} \ \frac{23}{70} \ \frac{31}{105} \ -\frac{11}{70}] = [0 \ 1 \ 0 \ 0 \ 0 \ | \ \frac{69}{280} \ \frac{27}{140} \ -\frac{23}{280} \ -\frac{127}{140} \ \frac{151}{280}] \\
& \rightarrow R_3 \rightarrow R_3 + \frac{15}{22}R_5 & \rightarrow R_4 \rightarrow R_4 - \frac{13}{132}R_5 \\
& = [0 \ 0 \ 1 \ 0 \ 0 \ | \ -\frac{3}{28} \ -\frac{3}{14} \ \frac{1}{28} \ \frac{11}{14} \ -\frac{9}{28}] = [0 \ 0 \ 0 \ 1 \ 0 \ | \ -\frac{33}{280} \ \frac{83}{420} \ \frac{11}{280} \ -\frac{43}{420} \ \frac{13}{280}]
\end{aligned}$$

$$\text{RREF} = \left[ \begin{array}{ccccc|ccccc}
1 & 0 & 0 & 0 & 0 & \frac{1}{70} & -\frac{11}{105} & \frac{23}{70} & \frac{31}{105} & -\frac{11}{70} \\
0 & 1 & 0 & 0 & 0 & \frac{69}{280} & \frac{27}{140} & -\frac{23}{280} & -\frac{127}{140} & \frac{151}{280} \\
0 & 0 & 1 & 0 & 0 & -\frac{3}{28} & -\frac{3}{14} & \frac{1}{28} & \frac{11}{14} & -\frac{9}{28} \\
0 & 0 & 0 & 1 & 0 & -\frac{33}{280} & \frac{83}{420} & \frac{11}{280} & -\frac{43}{420} & \frac{13}{280} \\
0 & 0 & 0 & 0 & 1 & \frac{3}{70} & -\frac{11}{35} & -\frac{1}{70} & \frac{31}{35} & -\frac{33}{70}
\end{array} \right]$$

$$A^{-1} = \left[ \begin{array}{ccccc}
\frac{1}{70} & -\frac{11}{105} & \frac{23}{70} & \frac{31}{105} & -\frac{11}{70} \\
\frac{69}{280} & \frac{27}{140} & -\frac{23}{280} & -\frac{127}{140} & \frac{151}{280} \\
-\frac{3}{28} & -\frac{3}{14} & \frac{1}{28} & \frac{11}{14} & -\frac{9}{28} \\
-\frac{33}{280} & \frac{83}{420} & \frac{11}{280} & -\frac{43}{420} & \frac{13}{280} \\
\frac{3}{70} & -\frac{11}{35} & -\frac{1}{70} & \frac{31}{35} & -\frac{33}{70}
\end{array} \right] = \frac{1}{840} \text{adj } A$$

2.

a.

$$\begin{aligned}
& R_2 \rightarrow R_2 - 2R_1 & \rightarrow R_3 \rightarrow R_3 - R_1 \\
& = [(2 - 2(1)) \ (4 - 2(3)) \ (-1 - 2(4))] = [0 \ -2 \ -9] = [(1 - 1) \ (3 - 3) \ (-6 - 4)] = [0 \ 0 \ -10]
\end{aligned}$$

$$\begin{aligned}
& \text{REF} = \left[ \begin{array}{ccc}
1 & 3 & 4 \\
0 & -2 & -9 \\
0 & 0 & -10
\end{array} \right] \quad (\text{row replacement does not change det}) \\
& \det B = (1)(-2)(-10)
\end{aligned}$$

$$\boxed{\det B = 20}$$

b.

$$\det B = a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13}$$

$$M_{11} = \begin{vmatrix} 4 & -1 \\ 3 & -6 \end{vmatrix} \rightarrow M_{12} = \begin{vmatrix} 2 & -1 \\ 1 & -6 \end{vmatrix} \rightarrow M_{13} = \begin{vmatrix} 2 & 4 \\ 1 & 3 \end{vmatrix}$$

$$= -24 + 3 = -21 \quad = -12 + 1 = -11 \quad = 6 - 4 = 2$$

$$\det B = 1(-21) - 3(-11) + 4(2) = -21 + 33 + 8$$

$$\boxed{\det B = 20}$$

c.

$$M_{11} = \begin{vmatrix} 4 & -1 \\ 3 & -6 \end{vmatrix} \rightarrow M_{12} = \begin{vmatrix} 2 & -1 \\ 1 & -6 \end{vmatrix} \rightarrow M_{13} = \begin{vmatrix} 2 & 4 \\ 1 & 3 \end{vmatrix} \rightarrow M_{21} = \begin{vmatrix} 3 & 4 \\ 3 & -6 \end{vmatrix}$$

$$= (4)(-6) - (-1)(3) = -21 \quad = (2)(-6) - (-1)(1) = -11 \quad = (2)(3) - (4)(1) = 2 \quad = (3)(-6) - (4)(3) = -30$$

$$\rightarrow M_{22} = \begin{vmatrix} 1 & 4 \\ 1 & -6 \end{vmatrix} \rightarrow M_{23} = \begin{vmatrix} 1 & 3 \\ 1 & 3 \end{vmatrix} \rightarrow M_{31} = \begin{vmatrix} 3 & 4 \\ 4 & -1 \end{vmatrix} \rightarrow M_{32} = \begin{vmatrix} 1 & 4 \\ 2 & -1 \end{vmatrix}$$

$$= (1)(-6) - (4)(1) = -10 \quad = (1)(3) - (3)(1) = 0 \quad = (3)(-1) - (4)(4) = -19 \quad = (1)(-1) - (4)(2) = -9$$

$$\rightarrow M_{33} = \begin{vmatrix} 1 & 3 \\ 2 & 4 \end{vmatrix}$$

$$= (1)(4) - (3)(2) = -2$$

$$C_{11} = +(-21) = -21, \quad C_{12} = -(-11) = 11, \quad C_{13} = +(2) = 2$$

$$C_{21} = -(-30) = 30, \quad C_{22} = +(-10) = -10, \quad C_{23} = -(0) = 0$$

$$C_{31} = +(-19) = -19, \quad C_{32} = -(-9) = 9, \quad C_{33} = +(-2) = -2$$

$$C = \begin{bmatrix} -21 & 11 & 2 \\ 30 & -10 & 0 \\ -19 & 9 & -2 \end{bmatrix} \rightarrow C^T$$

$$\boxed{\text{adj } B = \begin{bmatrix} -21 & 30 & -19 \\ 11 & -10 & 9 \\ 2 & 0 & -2 \end{bmatrix}}$$

d.

$$R_2 \rightarrow R_2 - 2R_1$$

$$= [(2 - 2(1)) (4 - 2(3)) (-1 - 2(4)) (0 - 2(1)) (1 - 2(0)) (0 - 2(0))] = [0 \quad -2 \quad -9 \quad | \quad -2 \quad 1 \quad 0]$$

$$\rightarrow R_3 \rightarrow R_3 - R_1$$

$$= [(1 - 1(1)) (3 - 1(3)) (-6 - 1(4)) (0 - 1(1)) (0 - 1(0)) (1 - 1(0))] = [0 \quad 0 \quad -10 \quad | \quad -1 \quad 0 \quad 1]$$

$$\rightarrow R_2 \rightarrow -\frac{1}{2}R_2$$

$$= [(-\frac{1}{2}(0)) (-\frac{1}{2}(-2)) (-\frac{1}{2}(-9)) (-\frac{1}{2}(-2)) (-\frac{1}{2}(1)) (-\frac{1}{2}(0))] = [0 \quad 1 \quad \frac{9}{2} \quad | \quad 1 \quad -\frac{1}{2} \quad 0]$$

$$\rightarrow R_1 \rightarrow R_1 - 3R_2$$

$$= [(1 - 3(0)) (3 - 3(1)) (4 - 3(\frac{9}{2})) (1 - 3(1)) (0 - 3(-\frac{1}{2})) (0 - 3(0))] = [1 \quad 0 \quad -\frac{19}{2} \quad | \quad -2 \quad \frac{3}{2} \quad 0]$$

$$\rightarrow R_3 \rightarrow -\frac{1}{10}R_3$$

$$= [(-\frac{1}{10}(0)) (-\frac{1}{10}(0)) (-\frac{1}{10}(-10)) (-\frac{1}{10}(-1)) (-\frac{1}{10}(0)) (-\frac{1}{10}(1))] = [0 \quad 0 \quad 1 \quad | \quad \frac{1}{10} \quad 0 \quad -\frac{1}{10}]$$

$$\rightarrow R_1 \rightarrow R_1 + \frac{19}{2}R_3$$

$$= [(1 + \frac{19}{2}(0)) (0 + \frac{19}{2}(0)) (-\frac{19}{2} + \frac{19}{2}(1)) (-2 + \frac{19}{2}(\frac{1}{10})) (\frac{3}{2} + \frac{19}{2}(0)) (0 + \frac{19}{2}(-\frac{1}{10}))] = [1 \quad 0 \quad 0 \quad | \quad -\frac{21}{20} \quad \frac{3}{2} \quad -\frac{19}{20}]$$

$$\rightarrow R_2 \rightarrow R_2 - \frac{9}{2}R_3$$

$$= [(0 - \frac{9}{2}(0)) (1 - \frac{9}{2}(0)) (\frac{9}{2} - \frac{9}{2}(1)) (1 - \frac{9}{2}(\frac{1}{10})) (-\frac{1}{2} - \frac{9}{2}(0)) (0 - \frac{9}{2}(-\frac{1}{10}))] = [0 \quad 1 \quad 0 \quad | \quad \frac{11}{20} \quad -\frac{1}{2} \quad \frac{9}{20}]$$

$$\text{RREF} = \left[ \begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{21}{20} & \frac{3}{2} & -\frac{19}{20} \\ 0 & 1 & 0 & \frac{11}{20} & -\frac{1}{2} & \frac{9}{20} \\ 0 & 0 & 1 & \frac{1}{10} & 0 & -\frac{1}{10} \end{array} \right]$$

$$B^{-1} = \begin{bmatrix} -\frac{21}{20} & \frac{3}{2} & -\frac{19}{20} \\ \frac{11}{20} & -\frac{1}{2} & \frac{9}{20} \\ \frac{1}{10} & 0 & -\frac{1}{10} \end{bmatrix}$$

$$\text{Check: } B^{-1} = \frac{1}{\det B} \text{adj } B = \frac{1}{20} \begin{bmatrix} -21 & 30 & -19 \\ 11 & -10 & 9 \\ 2 & 0 & -2 \end{bmatrix} = \begin{bmatrix} -\frac{21}{20} & \frac{3}{2} & -\frac{19}{20} \\ \frac{11}{20} & -\frac{1}{2} & \frac{9}{20} \\ \frac{1}{10} & 0 & -\frac{1}{10} \end{bmatrix} \checkmark$$